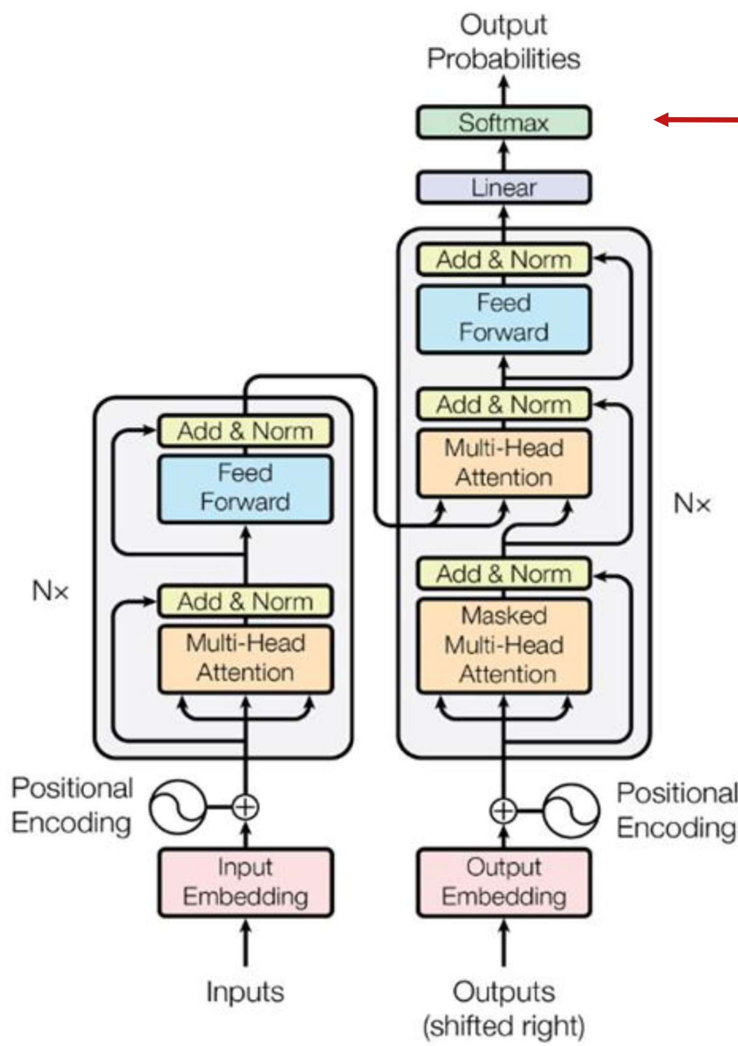
Transformer Architecture



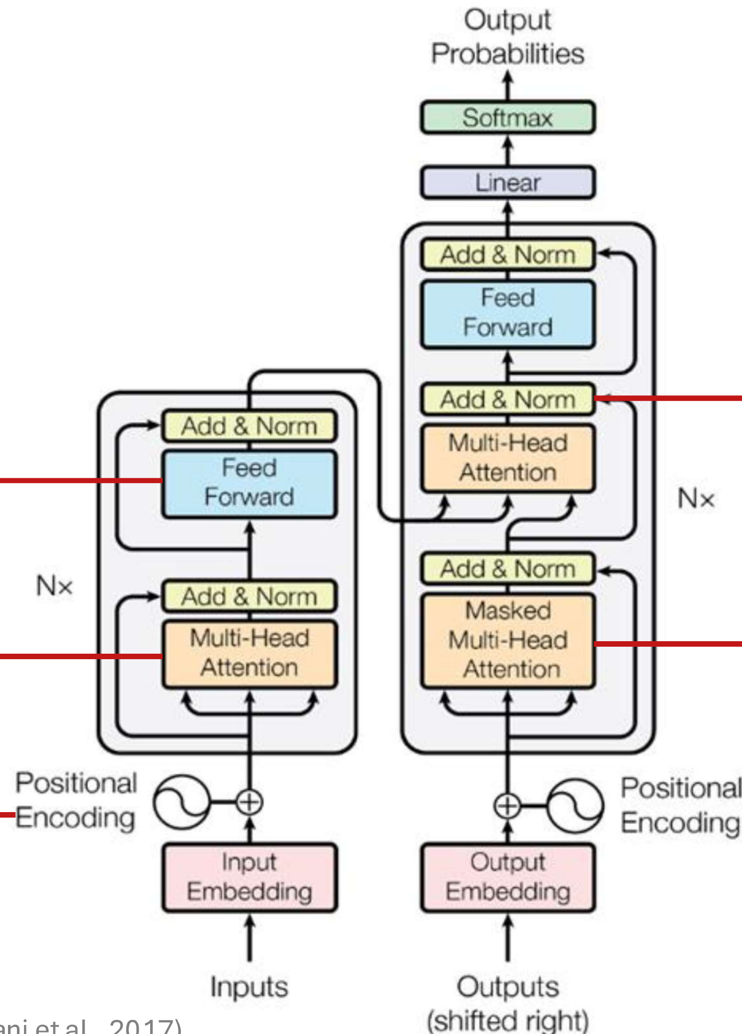
After stacking a bunch of these decoder blocks, we finally have our familiar **softmax** layer to predict the next English word.

Transformer Architecture

Adding non-linearities

Allows querying multiple positions at each layer

Adds positional information



Reduces covariance shift and makes the system stable

Prevents attention lookups into the future while decoding

Source of Image : Attention is all you need (Vaswani et al., 2017)

Layer normalization

- **Main idea:** Batch normalization is quite beneficial, but it's challenging to apply with sequence models. The varying lengths of sequences make it difficult to normalize across a batch. Sequences can be very long, which often results in smaller batch sizes.
- **Solution:** Layer normalization

d-dimensional vectors for each sample in batch

$a_1, a_2, \ldots, a_B$

d-dim

$$\mu = \frac{1}{B}\sum_{i=1}^{B} a_i \quad \sigma = \sqrt{\frac{1}{B}\sum_{i=1}^{B}(a_i - \mu)^2}$$

$$\bar{a}_i = \frac{a_i - \mu}{\sigma}\gamma + \beta$$

Batch Norm

different $dimensions$ of a

$$\mu = \frac{1}{d}\sum_{i=1}^{d} a_j \quad \sigma = \sqrt{\frac{1}{d}\sum_{i=1}^{d}(a_j - \mu)^2}$$

1-dim

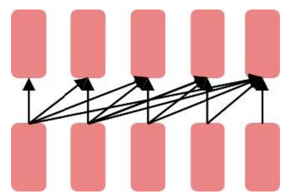
$$\bar{a} = \frac{a - \mu}{\sigma}\gamma + \beta$$

Layer Norm

Pre-training Strategies

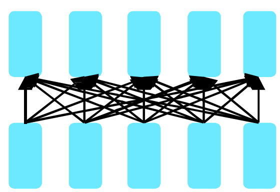
Pretraining for Three Types of Architectures

The neural architecture influences the type of pretraining, and natural use cases.



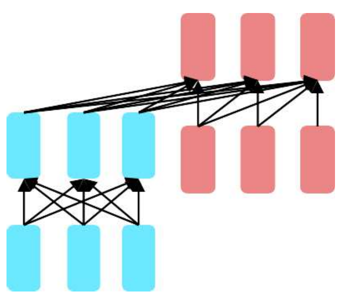
Decoders

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words



Encoders

- Gets bidirectional context – can condition on future!
- How do we pretrain them?



Encoder-Decoders

- Good parts of decoders and encoders?
- What's the best way to pretrain them?

BERT: Bidirectional Encoder Representations from Transformers

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Google AI Language
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Slides are adopted from Jacob Devlin

Background - Bidirectional Context

- Bidirectional context, unlike unidirectional context, takes into account both the left and right contexts.

Target Word

Apple is my favourite fruit and I eat it all the time.

Left Context

Right Context

Target Word

Apple is my favourite brand for buying laptop and other gadgets.

Left Context

Right Context

Motivation

- **Problem with previous methods:**
  - Language models only use left context or right context.
  - But language understanding is **bidirectional**.
- **Possible Issue:**
  - Directionality is needed to generate a well-formed probability distribution.
  - Words can see themselves in a bidirectional model.

Masked Language Modelling

10%

- Mask out k% of the input words, and then predict the masked words (Usually k = 15%). Example :

I like going to the [MASK] in the evening
↓
park

  - Too little masking: Too expensive to train
  - Too much masking: Not enough context
- The model needs to predict 15% of the words, but we don't replace with [MASK] 100% of the time. Instead:
  - 80% of the time, **replace with [MASK]**
    - Example : like going to the park → like going to the [MASK]
  - 10% of the time, **replace random word**
    - Example : like going to the park → like going to the store
  - 10% of the time, **keep same**
    - Example : like going to the park → like going to the park

Next Sentence Prediction

- To learn relationships between sentences, predict whether Sentence B is actual sentence that proceeds Sentence A, or a random sentence.

  Input = [CLS] I enjoy read [MASK] book ##s [SEP]
  I finish ##ed a [MASK] novel [SEP]
  Label = IsNext

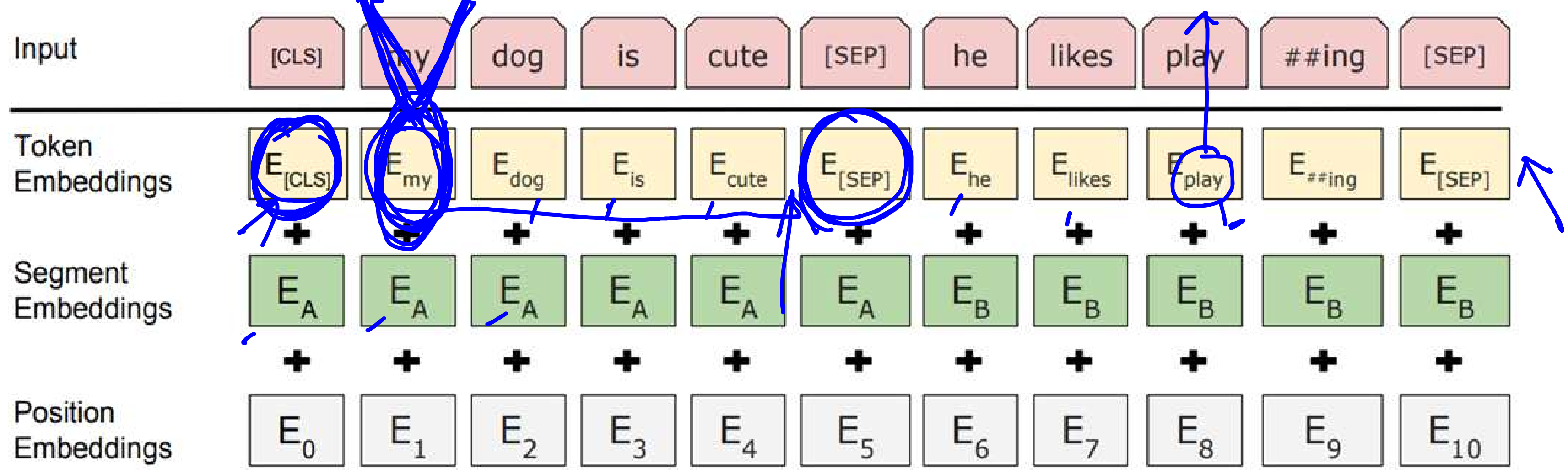
  Input = [CLS] I enjoy read ##ing book [MASK] [SEP]
  The dog ran [MASK] the street [SEP]
  Label = NotNext

- Important for many important downstream tasks such as Question Answering (QA) and Natural Language Inference (NLI)
- How to choose sentences A and B for pretraining?
  - 50% of the time B is the actual next sentence that follows A (labeled as IsNext)
  - 50% of the time it is a random sentence from the corpus (labeled as NotNext)

Input Representation

- Use 30,000 WordPiece vocabulary on input.
- For a given token, its input representation is constructed by summing the token embeddings, the segmentation embeddings and the position embeddings.

1) ✓
2) ✓



Source of Image : BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (Devlin et al., NAACL 2019)

Pre-Training Encoder-Decoder Models

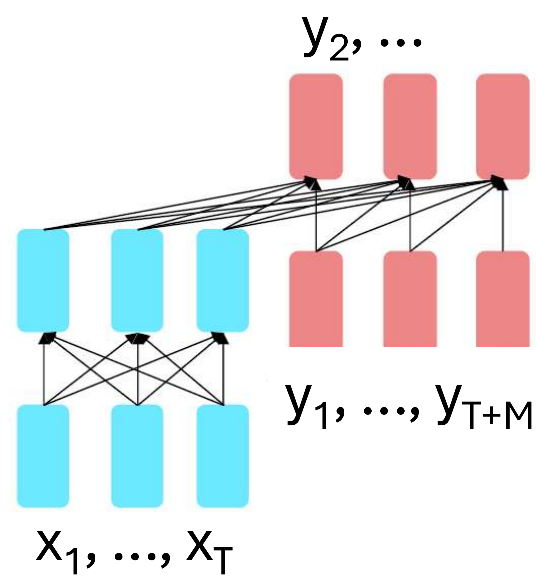
- For **encoder-decoders**, we could do something like **language modeling**, but where a prefix of every input is provided to the encoder and is not predicted.

$$h_1, \ldots, h_T = \mathit{Encoder}\,(x_1, \ldots, x_T)$$

$$h_{T+1}, \ldots, h_{T+M} = \mathit{Decoder}\,(y_1, \ldots, y_{i-1}, h_1, \ldots, h_T)$$

$$P(y_i \mid y_{<i}, h_{1:T}) = \mathit{Softmax}(Wh_i + b)$$

The **encoder** portion benefits from bidirectional context; the **decoder** portion is used to train the whole model through language modeling.



Pre-training

$\downarrow$

Fine tune

Pre-train $\rightarrow$ Fine tune $\rightarrow$ Predict

Post 2017.

2018 - BERT

GPT

GPT2 - 1.75B $\downarrow$ 110x

2018 $\downarrow$ 2020

2020 GPT3 175B

Pre-train $\rightarrow$ Prompt $\rightarrow$ Predict

BERT

Hotpot-QA

SRK

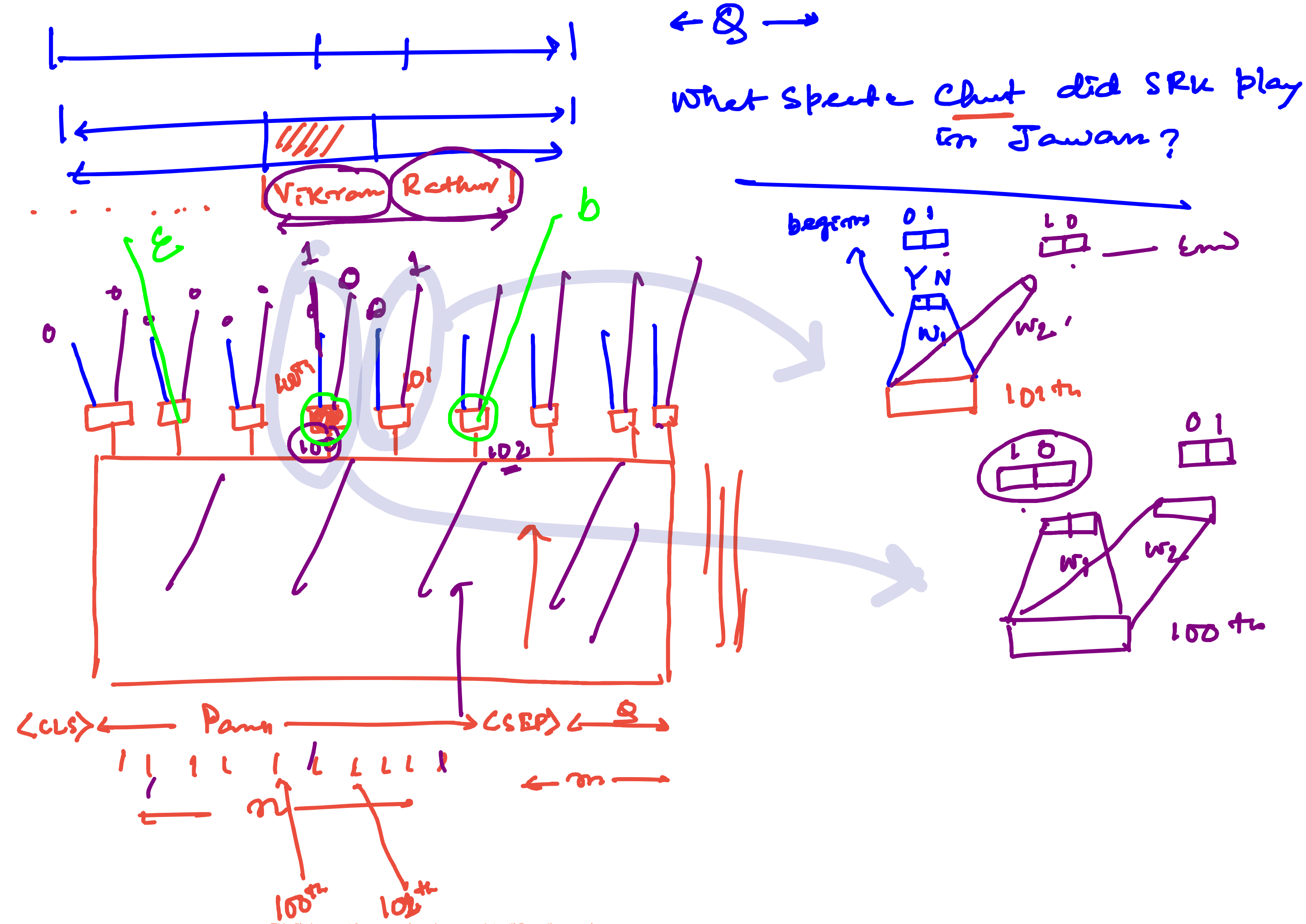
BI

1)

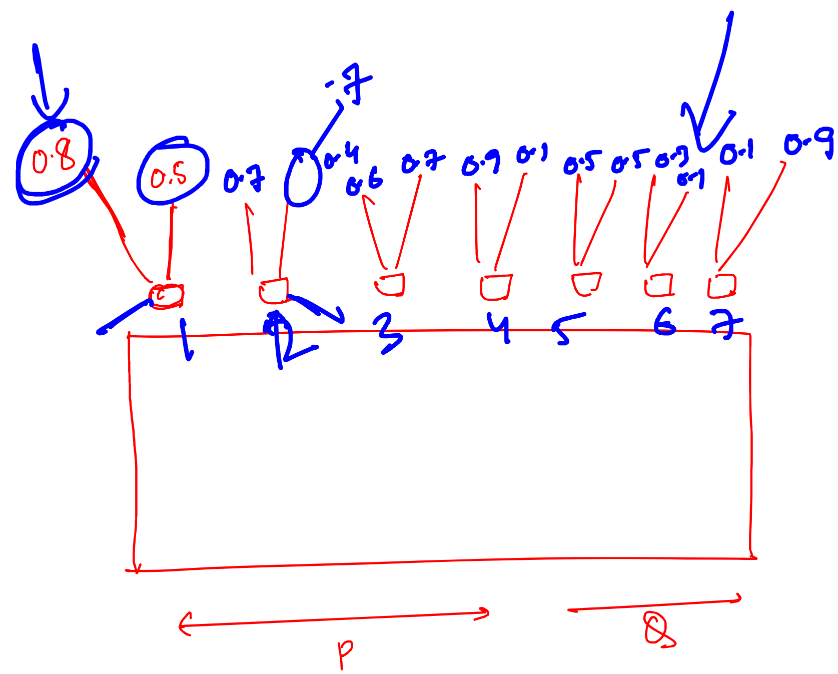
Q-A task | Closed book QA | Extractive QA

← Q →

What specific chat did SRK play in Jawan?



Ex T



Abstractive Summarization

$p_b \times p_a$

(2, 7)

$\underline{0.7 \times 0.9}$

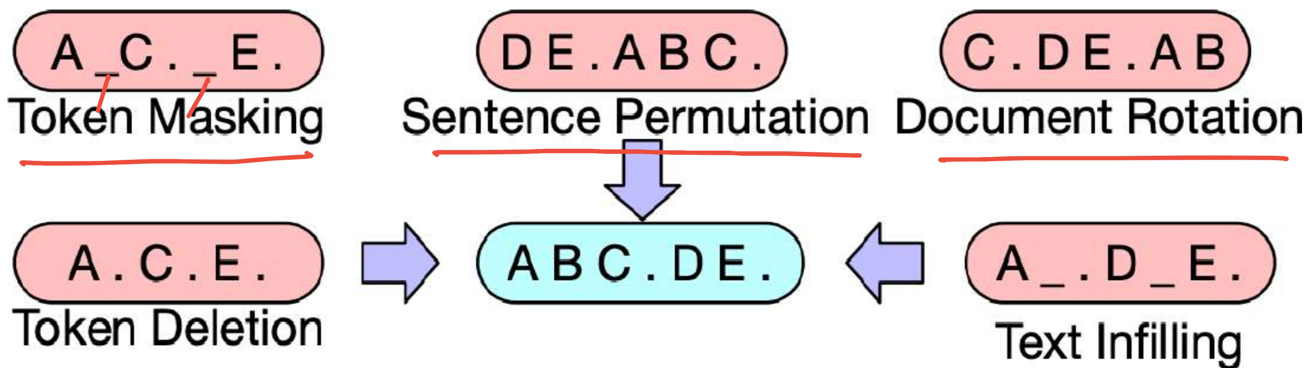
(3, 5)

$\underline{0.6, \times 0.5}$

Pre-Training Encoder-Decoder Models

- How can we pre-train a model for P(**y** | **x**)?
- **Requirements:**
  1. should use unlabeled data
  2. should force a model to attend from **y** back to **x**

Pre-Training BART (Bidirectional and Auto-Regressive Transformers)



Infilling is longer spans than masking

A B C D E

E A B C D

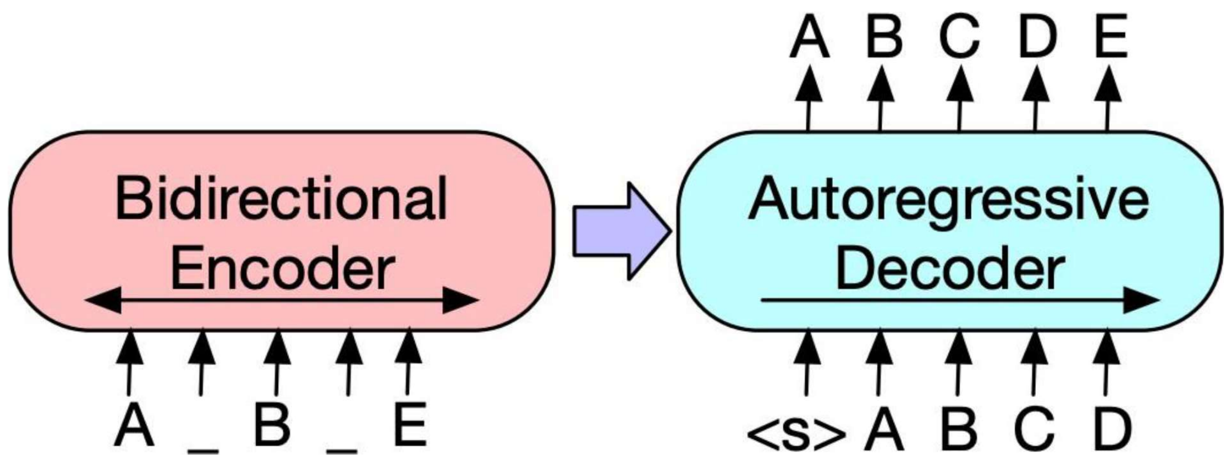
D E A B C

- Several possible strategies for corrupting a sequence are explored in the BART paper.

Lewis et al. (2019), "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension"

Pre-Training BART

- Sequence-to-sequence Transformer trained on this data: permute/make/delete tokens, then predict full sequence autoregressively.



Lewis et al. (2019), “BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension”

BART

GPT-3 2020

175 B

" Prompting "

Lan Lan n.

2020

2022

InstructGPT

RLHF * IRL

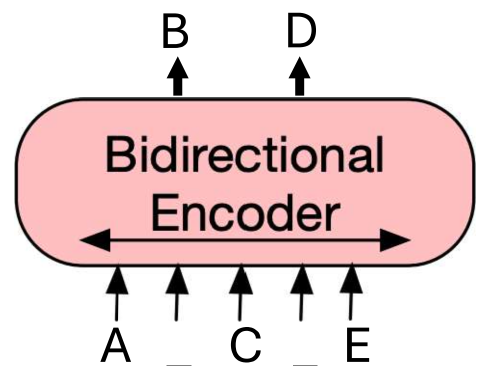
ChatGPT - Dec 2020

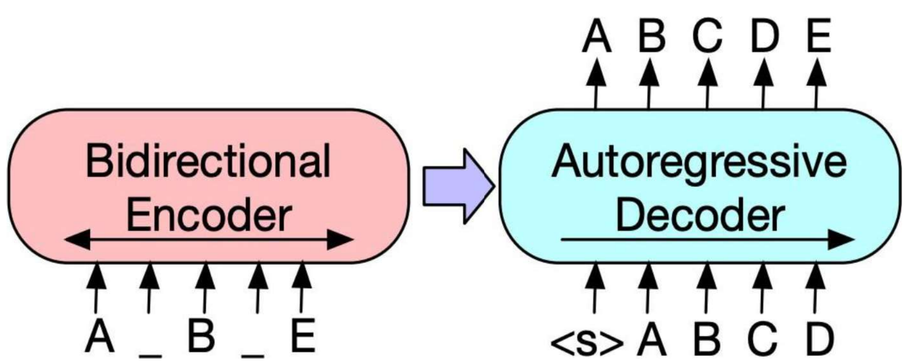
Reinforcement Learning From/with Human Feedback

Alignment

BERT vs. BART

- **BERT:** only an encoder, trained with masked language modeling objective. Cannot generate text or do Seq2Seq tasks (in standard form).





- **BART:** consists of both an encoder and a decoder. Can also use just the encoder wherever we would use BERT.

BART for Summarization

- **Pre-train** on the BART task: take random chunks of text, noise them according to the schemes described, and try to “decode” the clean text
- **Fine-tune** on a summarization dataset: a news article is the input and a summary of that article is the output (usually 1-3 sentences depending on the dataset)
- Can achieve good results even with **few summaries to fine-tune on**, compared to basic seq2seq models which require 100k+ examples to do well

BART for Summarization: Output Example

PG&E stated it scheduled the blackouts in response to forecasts for high winds amid dry conditions. The aim is to reduce the risk of wildfires. Nearly 800 thousand customers were scheduled to be affected by the shutoffs which were expected to last through at least midday tomorrow.

$\downarrow$

Power has been turned off to millions of customers in California as part of a power shutoff plan.

Lewis et al. (2019)

Add 2+2

'4'

C4

MC4

Token corruption

span

T5

Text to Text

Transfer Transform

<X> Very much <Y> inviting

Thank you <X> for <Y> time.

Thank you very much for inviting me.

<X>

<Y>

Add: 2+2

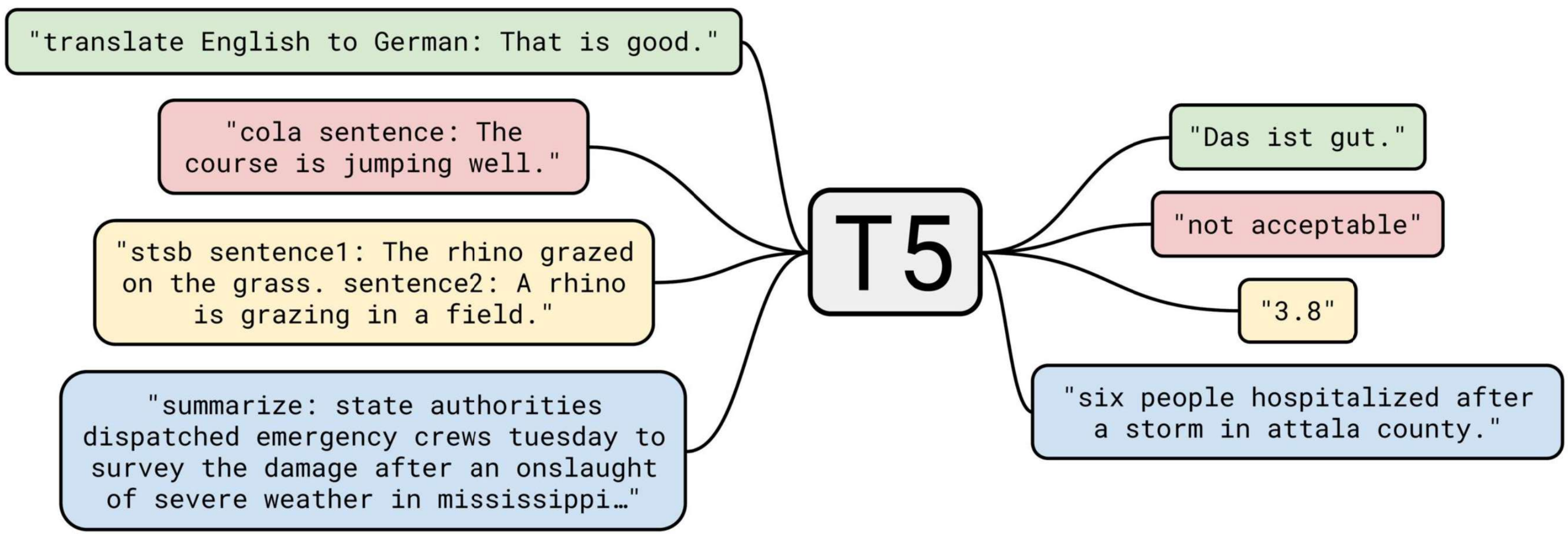
8:

<X>

Sentinal token

<Y>

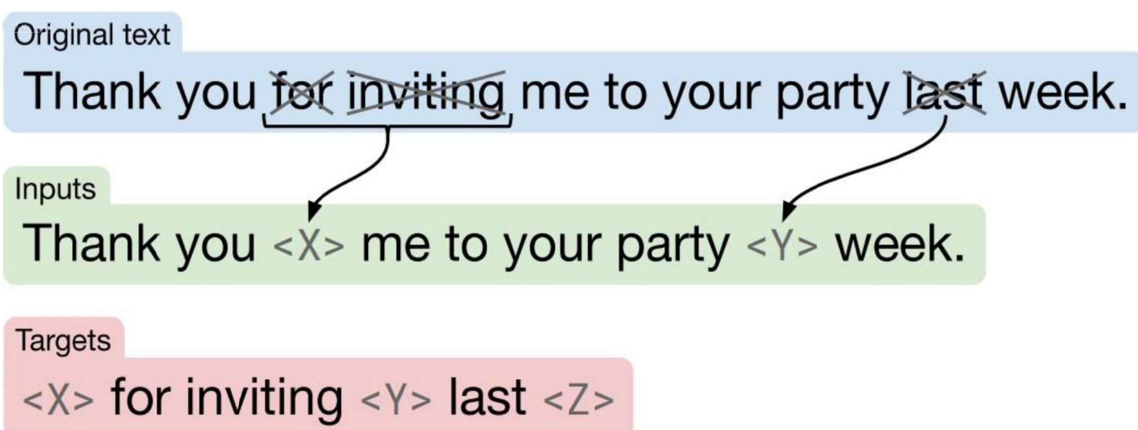
T5: **T**ext-**t**o-**T**ext **T**ransfer **T**ransformer



Raffel et al. (2019), "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer"

Pre-Training T5

- **Pre-training:** similar denoising scheme to BART (they were released within a week of each other in fall 2019)
- **Input:** text with gaps ; **Output:** a series of phrases to fill those gaps.



Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!

Raffel et al. (2019)

Pre-Training Decoder-only Models

GPT and Llama

Recall: Probabilistic Language Models

$$P(\underline{X})$$

Sentence/Document

A generative model that calculates the probability of language

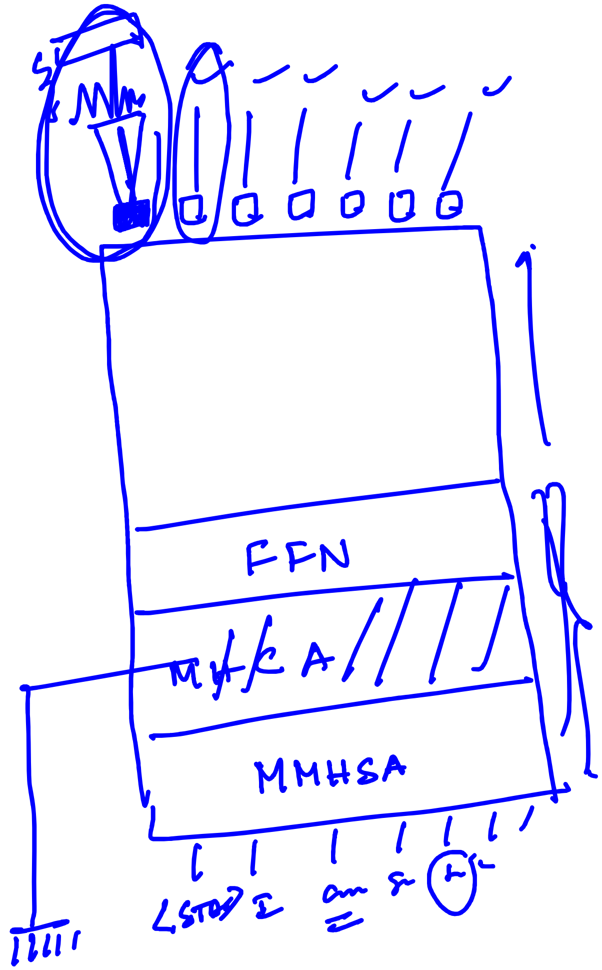
Auto-regressive Language Models

$$P(X) = \prod_{i=1}^{I} P(\underbrace{x_i}_{\text{Next Token}} \mid \underbrace{x_1, \ldots, x_{i-1}}_{\text{Context}})$$

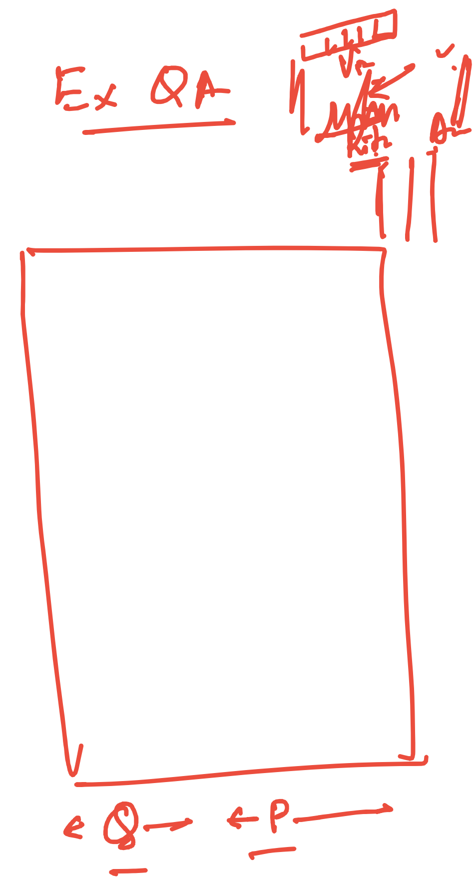
Pretraining Decoder only

GPT

Finetune it



Ex QA



Next Token Prediction

- This is essentially **classification**!
  - We can think of neural language models as neural classifiers. They classify prefix of a text into |V| classes, where the classes are vocabulary tokens.

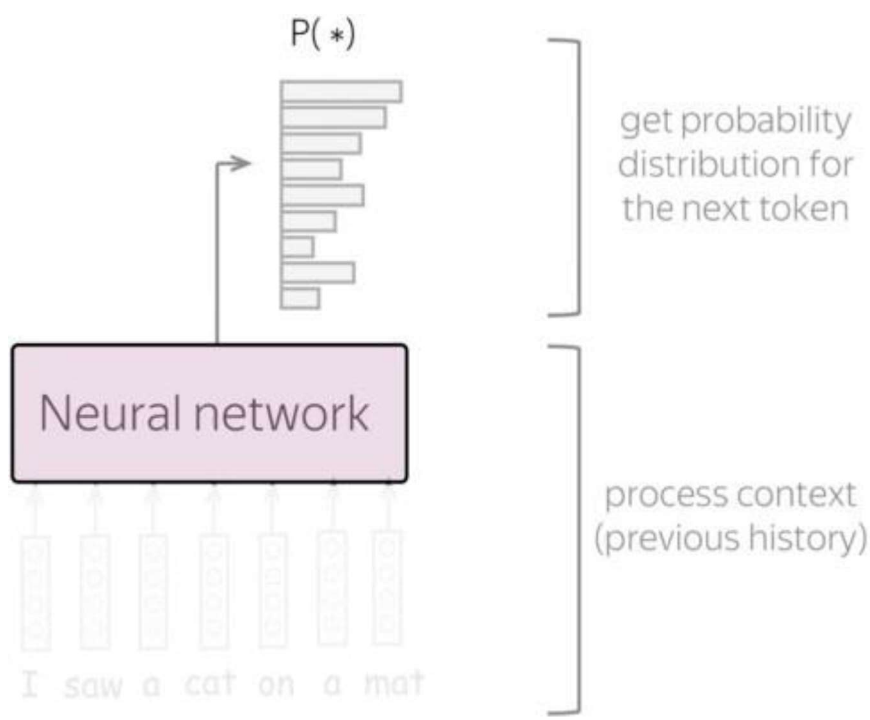
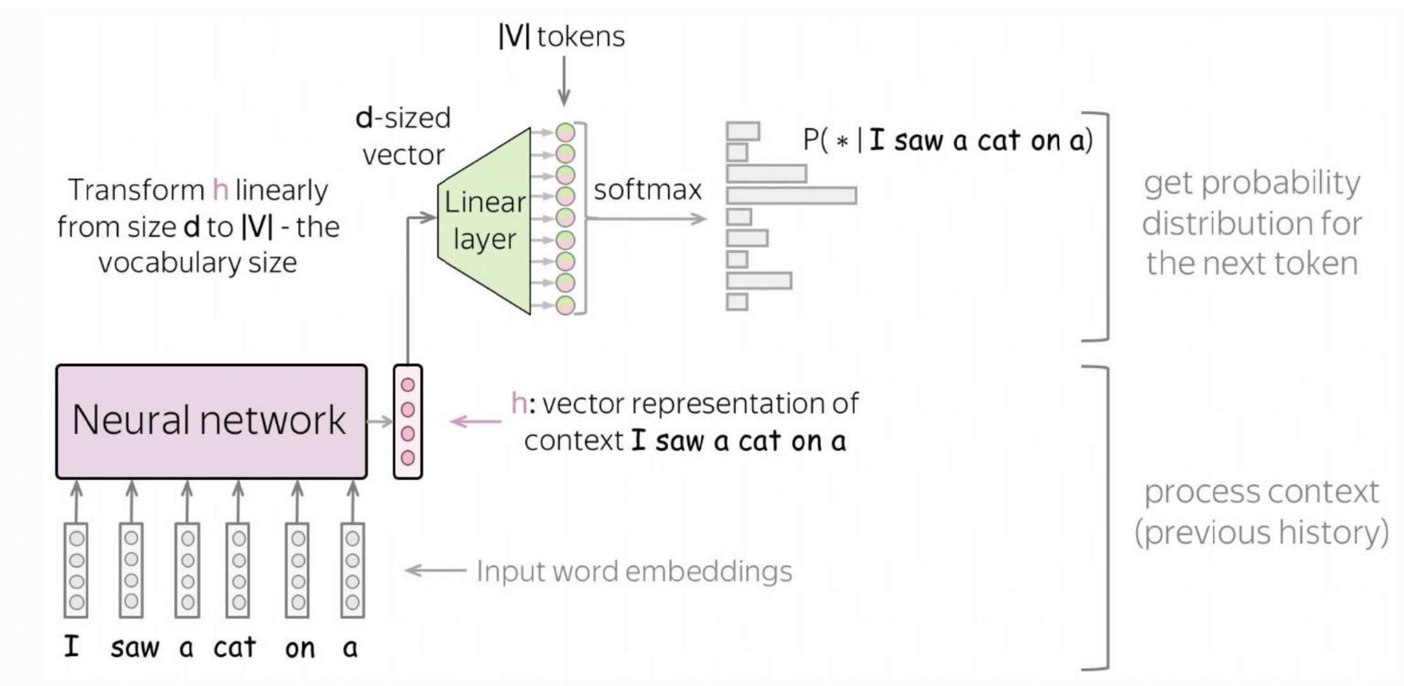


Image Credit: https://lena-voita.github.io/nlp_course/language_modeling.html

Next Token Prediction



- Feed word embedding for previous (context) words into a network.
- Get vector representation of context from the network.
- From this vector representation, predict a probability distribution for the next token.

Image Credit: https://lena-voita.github.io/nlp_course/language_modeling.html